Exercises 1A

1. Suppose $f : [a, b] \to \mathbb{R}$ is a bounded function such that $L(f, P, [a, b]) = U(f, P, [a, b])$ for some partition P of $[a, b]$. Prove that f is the constant function on $[a, b]$.

Proof. Let $a = x_0, \ldots, x_n = b$ denote the partition P of $[a, b]$. Since the lower and upper Riemann sums are equal, we have

$$\sum_{j=1}^{n}(x_j - x_{j-1}) \sup_{[x_{j-1},x_j]} f = \sum_{j=1}^{n}(x_j - x_{j-1}) \inf_{[x_{j-1},x_j]} f.$$

$$\implies \sum_{j=1}^{n}(x_j - x_{j-1})\Big(\sup_{[x_{j-1},x_j]} f - \inf_{[x_{j-1},x_j]} f\Big) = 0$$

The length of each interval is strictly positive, and $\sup_{[x_{j-1},x_j]} f \geq \inf_{[x_{j-1},x_j]} f$ for each j, so the above sum consists of only non-negative terms. Thus, we have $(x_j - x_{j-1})(\sup_{[x_{j-1},x_j]} f - \inf_{[x_{j-1},x_j]} f) = 0$ for each interval, and $(\sup_{[x_{j-1},x_j]} f = \inf_{[x_{j-1},x_j]} f)$. Thus, f is constant on every interval $[x_{j-1}, x_j]$.

$f(x) = f(a)$ for every $x \in [x_1, x_2]$. Suppose f is $f(a)$ on an interval $[x_{j-1}, x_j]$. Since f is also constant on the interval $[x_j, x_{j+1}]$, $f(x) = f(x_j) = f(a)$ for all $x \in [x_j, x_{j+1}]$. By mathematical induction, $f(x) = f(a)$ on every interval $[x_j, x_{j+1}]$. Therefore, f is constant on the entire domain. $\square$

2. Suppose that $a \leq s < t \leq b$. Define $f : [a, b] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } s < x < t, \\ 0 & \text{otherwise.} \end{cases}$$

Prove that f is Riemann integrable on $[a, b]$ and that $\int_a^b f = t - s$.

Proof. Consider the partition P of $[a, b]$ into $n \in \mathbb{N}$ intervals of equal length given by $x_i = a + i\frac{b-a}{n}$. Notice that

$$L(f, P, [a, b]) = \sum_{i=1}^{n} d \inf_{[x_{i-1},x_i]} f \geq (\lfloor \frac{n(t-s)}{b-a} \rfloor - 2)\frac{b-a}{n},$$

and similarly

$$U(f, P, [a, b]) = \sum_{i=1}^{n} d \sup_{[x_{i-1},x_i]} f \leq \lfloor \frac{n(t-s)}{b-a} \rfloor \frac{b-a}{n}.$$

Then,

$$L(f, [a, b]) \geq \sup_{n\in\mathbb{N}}(\lfloor \frac{n(t-s)}{b-a} \rfloor - 2)\frac{b-a}{n} = t - s = \inf_{n\in\mathbb{N}}(\lfloor \frac{n(t-s)}{b-a} \rfloor)\frac{b-a}{n} \geq U(f, [a, b]).$$

But, since $L(f, [a, b]) \leq U(f, [a, b])$ we have $L(f, [a, b]) = U(f, [a, b]) = t - s$, and hence $\int_a^b f = t - s$. $\square$

3. Suppose $f : [a, b] \to \mathbb{R}$ is a bounded function. Prove that f is Riemann integrable if and only if for each $\epsilon > 0$ there exists a partition P of $[a, b]$ such that

$$U(f, P, [a, b]) - L(f, P, [a, b]) < \epsilon.$$

Proof.

($\implies$) Suppose f is Riemann integrable, and let $I := \int_a^b f$. Let $\epsilon > 0$. Since $I = \sup_P L(f, P, [a, b]) = \inf_P U(f, P, [a, b])$, there is a partition P_1 so that $L(f, P_1, [a, b]) > I - \frac{\epsilon}{2}$ and a partition P_2 so that $U(f, P_2, [a, b]) < I + \frac{\epsilon}{2}$. Let P be the partition obtained by taking the union of P_1 and P_2. Since P is a finer partition than both P_1 and P_2, we have

$$L(f, P, [a, b]) \geq L(f, P_1, [a, b]) > I - \epsilon/2$$

and

$$U(f, P, [a, b]) \leq U(f, P_2, [a, b]) < I + \epsilon/2.$$

Hence, $|L(f, P[a, b]) - I| < \epsilon/2$ and $|U(f, P[a, b]) - I| < \epsilon/2$. By the triangle inequality, we have

$$\begin{aligned} U(f, P, [a, b]) - L(f, P, [a, b]) &=\leq |U(f, P, [a, b]) - L(f, P, [a, b])| \\ &= |U(f, P, [a, b]) - I + I - L(f, P, [a, b])| \\ &\leq |U(f, P, [a, b]) - I| + |I - L(f, P, [a, b])| \\ &< 2(\epsilon/2) = \epsilon, \end{aligned}$$

as desired.

($\impliedby$) Suppose that f is not Riemann integrable. Let $L := L(f, [a, b])$ and $U := U(f, [a, b])$ be the lower and upper Riemann integrals, where $L \neq U$. Since $L(f, P, [a, b]) \leq U(f, P, [a, b])$ for any partition, $L \leq U$. Let $\epsilon := \frac{U-L}{2}$. Then, for any partition P of $[a, b]$, we have

$$\begin{aligned} U(f, P, [a, b]) - L(f, P, [a, b]) &=\geq \inf_P U(f, P, [a, b]) - \sup_P L(f, P, [a, b]) \\ &= U - L \\ &> \epsilon. \end{aligned}$$

$\square$

4. Suppose $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable. Prove that $f + g$ is Riemann integrable and that $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.

Proof. Recall that the infimum and supremum distribute over the minkowski sums of sets. Then, we have

$$\begin{aligned}
L(f+g,[a,b]) &= \sup_P L(f+g,P,[a,b]) \\
&= \sup_P \sum_{j=1}^{n} (x_j - x_{j-1}) \inf_{[x_{j-1},x_j]} f+g \\
&= \sup_P \sum_{j=1}^{n} (x_j - x_{j-1}) \inf_{[x_{j-1},x_j]} f + \sup_P \sum_{j=1}^{n} (x_j - x_{j-1}) \inf_{[x_{j-1},x_j]} f \\
&= \int_a^b f + \int_a^b g \\
&= \inf_P \sum_{j=1}^{n} (x_j - x_{j-1}) \sup_{[x_{j-1},x_j]} f + \inf_P \sum_{j=1}^{n} (x_j - x_{j-1}) \sup_{[x_{j-1},x_j]} f \\
&= \inf_P \sum_{j=1}^{n} (x_j - x_{j-1}) \sup_{[x_{j-1},x_j]} f+g \\
&= \inf_P U(f+g,P,[a,b]) \\
&= U(f+g,[a,b])
\end{aligned}$$

Hence, $L(f+g,[a,b]) = U(f+g,[a,b]) = \int_a^b f + \int_a^b g$. □

5. Suppose $f : [a, b] \to \mathbb{R}$ is Riemann integrable. Prove that $-f$ is Riemann integrable and that $\int_a^b (f) = -\int_a^b f$.

Proof.

$$\begin{aligned}
L(-f,[a,b]) &= \sup_P \sum_{j=1}^{n} (x_j - x_{j-1}) \inf_{[x_{j-1},x_j]} (-f) \\
&= -\inf_P - \sum_{j=1}^{n} (x_j - x_{j-1})(-\sup_{[x_{j-1},x_j]} (f)) \\
&= -\inf_P \sum_{j=1}^{n} (x_j - x_{j-1})(\sup_{[x_{j-1},x_j]} (f)) \\
&= -\int_a^b f \\
&= -\sup_P \sum_{j=1}^{n} (x_j - x_{j-1})(\inf_{[x_{j-1},x_j]} (f)) \\
&= \inf_P - \sum_{j=1}^{n} (x_j - x_{j-1})(-\sup_{[x_{j-1},x_j]} (-f)) = U(-f,[a,b])
\end{aligned}$$

□

6. Suppose $f : [a,b] \to \mathbb{R}$ is Riemann integrable. Suppose $g : [a,b] \to \mathbb{R}$ is a function such that $g(x) = f(x)$ for all except finitely many $x \in [a,b]$. Prove that g is Riemann integrable on $[a,b]$ and

$$\int_a^b g = \int_a^b g.$$

Proof. Let $h := g(x) - f(x)$ be a function defined on $[a,b]$. Since $g = f + h$, and f is Riemann integrable, it suffices to show that h is Riemann integrable. Let $x_1, ..., x_n$ be the collection of all finitely many points so that $g(x_i) \neq f(x_i)$ for all i, given in increasing order. Moreover, let $C := \max_{i=1,\dots,n} |h(x_i)|$.

Let $\epsilon > 0$, and $N := \lceil \frac{b-a}{\epsilon} \rceil$. Then, let $P_1 := \{a = y_0, y_1, ..., y_N = b\}$ where $y_i = \min\{a + i\epsilon, b\}$, and let $P_2 := \{a, x_1, ..., x_n, b\}$. Consider the finer partition $P = P_1 \cup P_2$, whose elements shall be indexed $a = z_0, z_1, z_2, ..., z_M = b$. Since h is 0 almost everywhere, $\inf h = 0$ on every interval containing more than one point. Then, we have

$$\begin{aligned}
U(h, P, [a,b]) - L(h, P, [a,b]) &= \sum_{i=1}^{M} (z_i - z_{i-1}) \sup_{[z_{i-1}, z_i]} h - \sum_{i=1}^{M} (z_i - z_{i-1}) \inf_{[z_{i-1}, z_i]} h \\
&= \sum_{i=1}^{M} (z_i - z_{i-1}) \sup_{[z_{i-1}, z_i]} h \\
&\leq \sum_{i=1}^{M} \epsilon |\sup_{[z_{i-1}, z_i]} h| && \text{(by construction of } P_1\text{)}
\end{aligned}$$

Since h is 0 almost everywhere, $\sup h$ is 0 on every interval $[z_{i-1}, z_i]$ that does not contain some x_j. Let $m_1, ..., m_n$ be the sequence so that $z_{m_j} = x_j$ for every $1 \leq j \leq n$. Then,

$$\begin{aligned}
\sum_{i=1}^{M} \epsilon |\sup_{[z_{i-1}, z_i]} h| &\leq \sum_{j=1}^{n} \epsilon (|\sup_{[z_{m_j - 1}, z_{m_j}]} h| + |\sup_{[z_{m_j}, z_{m_j + 1}]} h|) \\
&\leq \epsilon \sum_{j=1}^{n} (|\sup_{[z_{m_j - 1}, z_{m_j}]} h| + |\sup_{[z_{m_j}, z_{m_j + 1}]} h|) \\
&\leq \epsilon \sum_{j=1}^{n} (2C) = 2Cn\epsilon
\end{aligned}$$

Since $2Cn$ is constant and ϵ was arbitrary, we have found a partition so that $U(h, P, [a,b]) - L(h, P[a,b]) \leq \epsilon$ for each ϵ. Hence, h is Riemann integrable, and consequently, so is g. □